The effect of deviations from the Kerr metric on the fluorescent Fe Kα line as a test for general relativity.

Student: Bradley Day,[1] Supervisor: Darius M. Michienzi[1]
[1]HH Wills Physics Laboratory, University of Bristol, Tyndall Avenue, Bristol BS8 1TL, UK

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This project investigated the the validity of the Kerr hypothesis and no hair theorem through deviations from the standard description of the spacetime around a black hole. The investigation was conducted using the Johannsen metric Johannsen (2013) via a table model of emission line profiles generated using the relativistic raytracing code `Gradus.jl` Baker & Young (2025b).

1 BACKGROUND

Emission line fitting is a useful tool in the study of compact objects due to the relativistic broadening of emission lines. The Fe Kα line, caused by the illumination of infalling matter by a hot corona above a black hole (BH), is a prominent feature of the spectra of BHs Fabian et al. (2000).

The Johannsen metric allows for deviations from the standard Kerr description of a rotating black hole via the deformation parameters α_{13} and ϵ_3. These parameters have both independently been used to conduct strong field tests of general relativity through emission line fitting Bambi et al. (2017); Cao et al. (2018); Tripathi et al. (2019, 2026), however they have not been simultaneously constrained, which was the goal of this project.

2 THE PARAMETER SPACE OF THE JOHANNSEN METRIC.

`Gradus.jl` Baker & Young (2025b) was used to explore the parameter space of the Johannsen metric by computing the observed emission line profile, and rendering images of the accretion disk to observe the redshift. It was found that the radius of the innermost stable circular orbit (ISCO) increases for increasing α_{13} and decreases for increasing ϵ_3. The orbital frequency for a test particle increases for increasing α_{13} leading to an increase in the degree of observed redshifting Johannsen (2013). The opposite occurs for ϵ_3. For $\alpha_{13} \lesssim 5.7$ the behaviour of the metric becomes unpredictable due to the presence of two local minima in the potential of an orbiting particle, and the energy and orbital frequency becoming imaginary Johannsen (2013).

Johannsen defined a lower limit for both of these parameters in equations 75,77 of Johannsen (2013). Below this limit there exist singularities or time-like curves in the exterior region of the black hole. Experimentation showed four additional unphysical regions of the Johannsen parameter space that exist only when α_{13} and ϵ_3 are varied simultaneously. These regions yielded either naked singularities or no solution for the ISCO.

These regions were found to take triangular shapes in the 2D plane created by the two deformation parameters, and could thus be bounded by linear equations. The shape of these regions varies with the spin parameter, a_*, thus a table was created of slopes and intercepts as a function of a_* to define exclusion regions in the parameter space.

3 FITTING TO OBSERVATIONS

A five-dimensional data table was constructed using `Gradus.jl` by varying the parameters a_*, h, θ, α_{13}, and ϵ_3, being the spin, corona height, inclination, and two deformation parameters, respectively. The grid size along each dimension was 10, ?, 9, 9, 10, respectively. For parameter combinations in the defined exclusion regions, an algorithm was defined to find the nearest "safe" combination in parameter space, and the line profile was calculated for this combination in place of the invalid one.

Data was used from the XRISM observatory XRISM Science Team (2020). [ENTER DETAILS HERE].

Fitting was completed using `SpectralFitting.jl` Baker & Young (2025a), the fit model was defined as

```
Abs * (PowerLaw + Conv(XILLVERD5))
```

where `Abs` models photoelectric absorption, `PowerLaw` describes the underlying power law continuum, and `XILLVER` models the reflection spectrum from the accretion disk García & Kallman (2010); García et al. (2013, 2014). The reflection spectrum was convolved by the table model defined above to give the relativistic spectrum.

- Fitting results

4 CONCLUSIONS

[ENTER CONCLUSIONS HERE]

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DATA AVAILABILITY

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